Question 1.1.2 While Euclidean distance is a more commonly known method to measure distance, it's not the only one! Another method is the Manhattan distance. How do we calculate the Manhattan distance between two points? What situation might Manhattan distance be more suitable than Euclidean distance? **(4 points)**

Hint: Consider why it's named *Manhattan distance*—think of a city grid!

$|x1-x2|+|y1-y2|$. Manhattan distance is more suitable than Euclidean distance in situations where movement is restricted to grid-like paths, where you can't only move diagonally.

Question 1.7.1 When doing Knn classification we split our data into training and test sets.

Why do we divide our data into training and test sets? Or in other words what is the point of the training set? What is the point of the test set? Answer both questions. **(7 points)**

> *Hint:* Check out this section in the textbook.

The training set helps the model learn and recognize patterns from the data. Test set is used for evaluating how well the train model performs on new, unseen data

Question 1.7.2 Why do we only want to use the test set once? **(3 points)**

We should only evaluate the test set a single time since it is meant to represent new, real-world data that the model has never seen.

Question 1.8. Why do we choose `k` to be an odd number in k-NN? Explain. **(10 points)**

We pick an odd k to avoid ties when choosing the most common class.